

Summary

About the Role:

The Senior Developer will support the modernization of federal statistical systems with AWS-native application development in Python and Java. Responsibilities include designing and maintaining backend services, translating legacy logic into modern architectures, writing complex SQL, supporting production workflows, and documenting technical processes. This is a fully remote role with occasional in-person meetings in Bethesda, MD, and requires a dependable developer who communicates clearly and thrives in a distributed environment.

Key Responsibilities:

- Develop backend services and containerized APIs using Python within AWS (Lambda, ECS/EKS, or similar).
- Build and maintain AWS-native components including Lambda, S3, RDS, API Gateway, Step Functions, and EventBridge/CloudWatch.
- Develop and support backend components written in Java.
- Write and optimize complex SQL queries for relational databases.
- Analyze legacy SAS-based processing logic and numerical/statistical algorithms to support modernization and knowledge transfer.
- Document workflows, algorithms, and system behavior using flowcharts (Visio or similar) and clear pseudo code.
- Support production operations, including scheduled processing, monitoring, log troubleshooting, and resolving backend issues.
- Collaborate with senior staff, statisticians, and SAS developers to understand backend processes, algorithms, and dependencies.
- Participate in Agile routines, code reviews, design discussions, and provide consistent communication in a remote environment.

Requirements

- 5+ years Python development experience (backend systems, APIs, data processing).

- 4+ years hands-on AWS development experience with serverless and/or containerized services.
- 4+ years Java development experience.
- 3+ years experience with relational databases and complex SQL.
- Experience with Visio or similar tools for system diagrams and flowcharts.
- Ability to write structured pseudo code describing algorithms and business rules.
- Experience developing or supporting containerized applications (Docker) preferred.
- Experience analyzing or implementing mathematical/statistical algorithms preferred.
- Strong analytical and troubleshooting skills with experience supporting production environments.
- Effective communication and reliability in a fully remote environment.
- Ability to review legacy logic and translate it into modern, maintainable solutions.